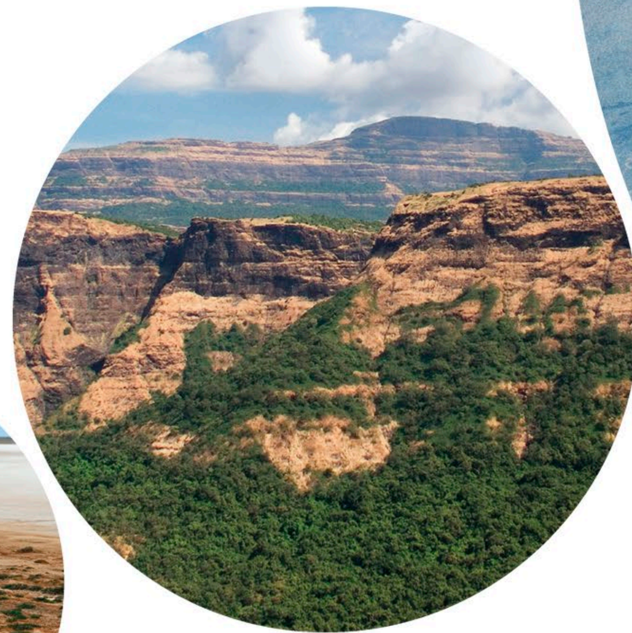
▷ **WORN HILLS**
The Urals are one of the world's oldest mountain ranges, although they have been worn down from their original heights by millions of years of weathering and erosion.

▷ **MOLTEN REMNANTS**
Rocks left over from the basalt lava eruptions that formed the Deccan Traps 66 million years ago have been exposed by the erosion of less resilient rocks around them.



◁ **ANCIENT LAND**
The ancient rock beneath the Kazakh steppe was once an entirely separate island continent geologists call Kazakhstania. Now, it is entirely trapped deep inside Asia.

THE SHAPING OF ASIA

Asia is bigger than any other continent and contains the world's highest mountain and largest mountain range. Yet it is also a young continent, formed from a complicated assembly of fragments.

The giant continent

All the continents have joined each other and split apart again and again—except Asia. It has stayed largely intact throughout Earth's history and is a recent creation, an amalgam of many pieces. The vast slab of ancient rock beneath Siberia, known as the Siberian, or Angaran, Platform, forms the continent's core, but it was joined by the other parts relatively recently. Even when the Siberian Platform was part of the supercontinent Pangaea 200 million years ago, south and east Asia were an assortment of separate island continents.

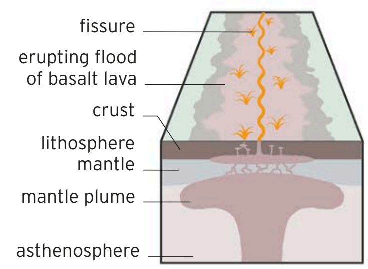
Between 100 and 50 million years ago, these fragments began to come together rapidly. First, north and south China and Southeast Asia fused onto Eurasia's southeast corner. Then, far to the south, India's extraordinary journey began. About 80 million years ago, India was attached to southern Africa, but then it broke away and hurtled north across an ancient ocean and smashed into the southern edge of Eurasia 60–40 million years ago, helping cement all the other fragments in place. Asia as we know it was born.

Volcanic India

India's Deccan Plateau is dominated by one of the largest volcanic features in the world, the Deccan Traps. The traps are what geologists call a large igneous province and are made of layers of volcanic basalt over 6,600 ft (2,000 m) thick. Even now, they spread over an area of 193,000 square miles (500,000 square km), but before being worn away by erosion they covered half of India.

Unlike some lavas, basalt is very runny and can flood onto the surface in huge quantities. The eruptions began 66.5 million years ago, when India had only recently become detached from Africa. One theory suggests that at that time India was above a hotspot in Earth's interior where the island of Réunion is now. At a hotspot, a fountain of hot magma

CREATION OF THE DECCAN TRAPS
Basalt lava erupted via a fissure in Earth's crust above a mantle plume, a fountain of hot rock rising in the mantle. The lava hardened and formed the Deccan Traps.



KEY EVENTS

252–250 million years ago Intense volcanic activity due to a mantle plume causes eruptions over a large area that will become eastern Siberia, creating a huge plain of volcanic rock called the Siberian Traps that still exists today.

130 million years ago A rain forest forms in an area that will become Malaysia. A portion of this survives in Taman Negara National Park today.

66 million years ago Huge volcanic eruptions on a landmass that will become India occur. Lava spreads and solidifies into the Deccan Traps.

60–40 million years ago The Indian Plate crashes into the southern part of the Eurasian Plate, causing the crust to crumple into the Himalayas.

50–40 million years ago The rise of the Himalayas blocks rainfall heading north of the new mountain range, causing the area to dry out and become the Gobi Desert.

5.5 million years ago An ancient sea that had spread from eastern Europe to western Asia dries out as land lifts and sea levels drop, leaving behind the Caspian, Aral, and Black Seas and Lake Urmia.

11,000 years ago Sumatra, Java, and Borneo become islands as sea levels rise.